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1 SUPER !!

lps Mar 26

Food + Oxygen $\longrightarrow$ Carbon dioxide + energy + water

VI
VII

3

3

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Mitochondrien

3

- (e) Design an experiment to show that living organisms respire.

Diagram	Method
	Water is heated to around 37°C in a beaker, checking using a thermometer. A glucose solution is put in a test tube. Yeast is put in too. The test tube is put in the beaker. Over time, the yeast will respire and bubbles form will appear on top.

12

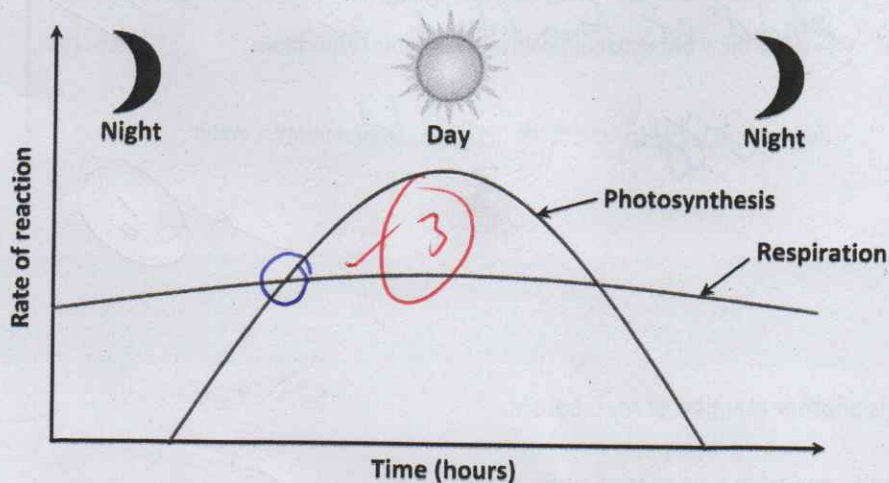
- (f) Describe the relationship between the respiratory system and respiration.

The respiratory system is the system that delivers all necessary ingredients for respiration to the cell and disposes of the waste products.

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Question 3

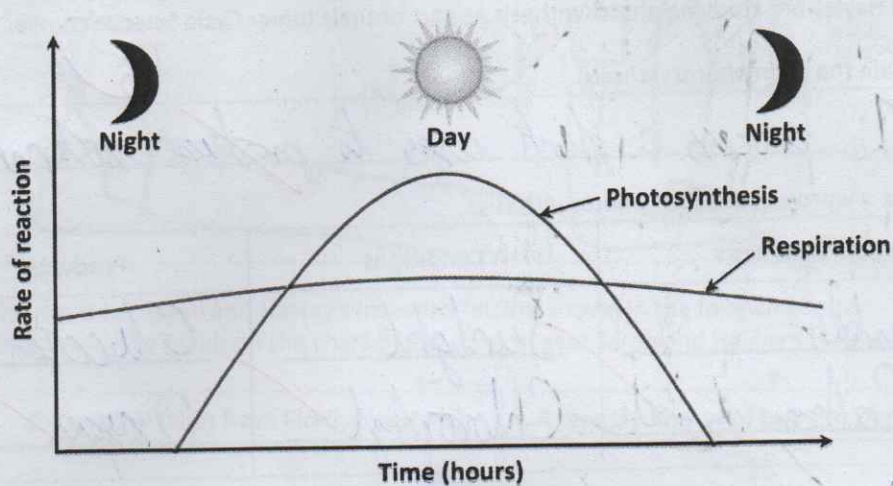
- (f) The graph below shows how the rates of photosynthesis and respiration for a plant varied over a period of 24 hours.



- (i) On the graph above, draw a circle around a point where the rate of respiration equals the rate of photosynthesis.

Question 4

- (f) The graph below shows how the rates of photosynthesis and respiration for a plant varied over a period of 24 hours.



- (iii) Explain why the rate of respiration remains relatively constant over 24 hours, compared to the rate of photosynthesis.

Photosynthesis requires constant, fresh daylight to function, whereas respiration can happen at any time of day

✓ (5)

Question 5

Question 12

(45 marks)

Sarah and Hayley are studying photosynthesis as part of their Junior Cycle Science course.

- (a) Explain the term *photosynthesis*.

The process a plant uses to produce substance

- (b) Write a word equation for photosynthesis.

Reactants	Conditions	Products
Water + Carbon dioxide	Sunlight → Chlorophyll	Glucose + Oxygen

- (c) Explain what is meant by an *autotroph*.

- (d) Sarah and Hayley's teacher gave them the following equipment to investigate if light intensity has an effect on the rate of photosynthesis: **desk lamp, thermometer, test tube, beaker, scalpel, Elodea (pondweed), metre stick, stopwatch**.

Explain how Sarah and Hayley would carry out the procedure for this investigation, using only the equipment listed above.

The elodea is placed in the beaker which is filled with water. A test tube is placed upside down over it. A thermometer is placed in the beaker. The metre stick is used to measure a specific distance for the lamp away from the beaker. A stopwatch is started and they

- (e) State two safety precautions that Hayley and Sarah will have to consider for this investigation.

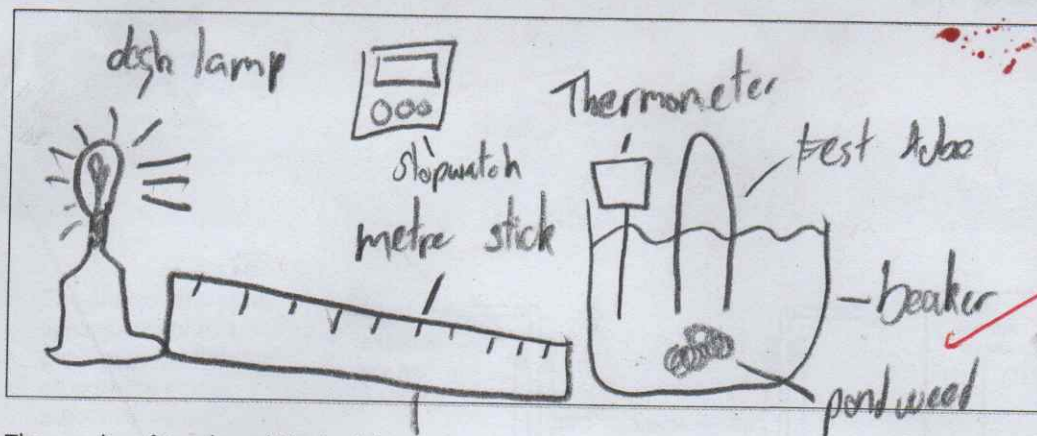
Keep away from electrical outputs
Tie back hair so as not to interfere

- (f) Carbon dioxide levels affect photosynthesis. Name two other factors that affect photosynthesis.

Light intensity
Temperature

count the bubbles that appear in the test tube

- (g) Draw a labelled diagram of the set-up of this experiment.



- (h) The results of Sarah and Hayley's investigation are shown in the following table. Draw a suitable graph on the chart below to represent Sarah and Hayley's results.

Distance of lamp from Elodea (metres)	Average number of bubbles per minute
1	10
0.75	19
0.5	24
0.25	30
0.1	33

